User Interface (UI) Basics

Duke OLLI Digital Explorers SIG

David Shamlin March 2026

Description of UI in the broadest and simplest terms

These are the key/core concepts of working with device/app UIs. Understanding these concepts and keeping them "front of mind" when using your device should make using your device(s) API much more meaningful (i.e., "intuitive").

- When interacting with a devices UI, we are performing **actions** on **objects**.
    - An **object** is anything I can see on my screen that I can interact with using my mouse/trackpad/touchscreen.
    - An **action** is an operation the can be performed on/by/with an **object**.
    - E.g., When I can print a file I create with apps like Pages or Word, printing is the action and the object is the file.
    - **Actions** are like verbs in a sentence and **objects** are like the subject and/or object of a sentence.
- When performing an **action**, we can typically/frequently also provide some additional information to tell the device how we want that **action** performed. E.g.,
    - When I save a file, I can tell my device the name I want the file to have
    - When printing a file, I can tell my device how I want my device to orient the file on paper; i.e., portrait or landscape
- This kind of information additional information is called a **parameter**.
- When an action has associated parameters, a special class of **objects** are used to allow you to give your preferred values for parameters. Objects of this class are called **controls**. Textboxes, checkboxes, and buttons are examples of **controls**.
- If/when there are **parameters** associated with an **action** and I don't explicitly give my device a value for on or more of the **action**'s **parameter**, the device will use some value predetermined by the software engineer(s) that created the app I am using. These predetermined values are called the **default** parameter values.

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Special Types of Objects

- **Window**
  - Panes: a window may be divided into sections; each section is referred to as a pane. Panes can often be hidden/revealed via a menu action or icon found in the window
- **Dialog**: a subtype of windows typically used to modify an actions parameters
- **Menu**: a collection of related actions
  - **Menu bar:** the collection of menus at the top left hand side of the screen that
  - **Pop-up menu** (aka, context menu): a menu that appears on screen when you click an icon
  - **Context menu**: a menu that appears when you ALT-CLICK (see section on Gestures) on certain locations on the screen
- Controls: buttons, checkboxes, textboxes, etc (see section on Controls)
- The UI treats the following as objects in the sense that you can interact with them using your mouse/trackpad/touchscreen to perform actions on them.
  - Text
  - Images
  - Files
  - Apps

Window & Control States

- Similarly to app, windows & controls also have states, and the UI provides you with visual “clues” that indicate a window or control’s current state.
- Window states
  - **Foreground**: when a window is in the foreground, text and icons in its header will be black (as opposed to gray)
  - **Background**: when a window is in the background
    - Text and icons in its header will be gray (as opposed to black)
    - One or more other windows may appear “over” or “on top of” a window in the background

  The above two states are mutually exclusive; i.e., a window is either in the foreground or in the background at any given moment.

  - Only one window is in the **foreground** at a time.
  - When a window is in the foreground, it is also said to be **in focus** and any keystrokes you make on keyboard are “sent to” that window.
  - Any pointing devices gestures you make on a window that is in the **background** causes that window to move to the foreground (and the window that was in the foreground will move to the background).
- Control states
  - **Disabled** (aka **inactive**): when a control is in this state, it will be gray and nothing will happen if you attempt to interact with it using your pointing device. When a control is in this state, there will often be some other control(s) in the same window that you need to interact with first; once you have done that, the disabled control will become enabled/active and you can then interact with it.
  - **Enabled** (aka **active**): when a control is in this state, it will be in some other color(s) besides gray and you can readily interact with it.

  The above two states are mutually exclusive; i.e., a control is either enabled/active or disabled/inactive at any given moment.

  - **In focus**: When a control is **enabled**, it may also be **in focus**. When a control is **in focus**, it will immediately “receive” any keystrokes you make using the keyboard. Usually a control will be highlighted when it is **in focus**; buttons are sometimes the exception. I.e., a button may be in focus without being highlighted. In some dialog windows that have more than one button to “complete”/“dismiss” the dialog (e.g., “OK” and “Cancel”, “Save” and “Cancel”, “Sign on” and “Cancel”, etc), one of the buttons may be in color—or in a brighter/darker color than the other. When this is the case, hitting the keyboard’ return key is the same as clicking on the button with your pointing device.

Software States

- At a given moment in time, an app is in in a specific state.
  - Analogy: Imagine a TV. At any moment in time, the TV can be either on or off. “On” and “off” are two states the TV can be in. When the TV is on, it is tuned to a specific channel; the channel the TV is turned to is another state of the TV.
- The basic states of apps are
  1. **Running** (aka, “open”, “executing”)
  2. **Shutdown** (aka, “closed”)

     The above two states are mutually exclusive. I.e., an app is either running or shutdown at a given moment in time.

  3. **Foreground** (aka, “active”, “in focus”)
  4. **Background** (aka, “suspended”, “asleep”)

     An app is one of the two above states when it running. I.e., when an app is running, it is either running in the foreground or in the background at a given moment in time. These two states are also mutually exclusive. I.e., a running app is either in the foreground or in the background but not both in the foreground and the background.
- Your device’s UI provides visual “clues” to the state(s) an app is in (so familiarity with these states should be helpful particularly when you have more than one app running at a time.)

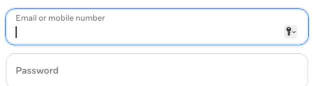
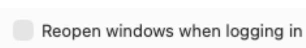
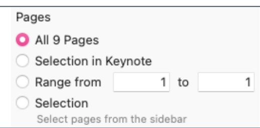
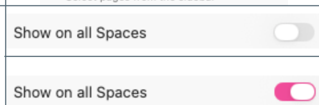
The Menu Bar

- Each item in the **menu bar** is called a **menu**.
- The first menu in the menu bar is always by the Apple menu ()—aka, “system menu”. The second menu is called the **app menu**, and it will always have the name of the app that you are currently using.
    - Note: looking at the name in the app menu is an easy/convenient way to determine which app is **active/ in the foreground**.
- Clicking on the name of a menu causes that menu’s content to appear.
- Each item in a menu’s content is called and **action** or **command**. Clicking on an action/command causes the app to perform—or **execute**—that action/command.
- When you see an action/command in one app that you have also seen and know how to use in another app, you can be highly confident that action/command behaves the same way in both apps.
- macOS menus adhere to a very consistent pattern; i.e., you will notice many similarities in menus across apps. E.g., all apps that allow you to create files will have New, Open, Save, and Save as… action/ command items in the Files menu.

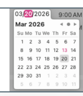
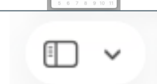
Standard macOS Menus

Name	Description
	Called the “system menu” (or “Apple menu”). This menu never contains actions related to the app you are using; it always/only contains actions that pertain to the device “at large”/“in general.”
app	Contains actions that apply to the app as a whole, rather than to a specific task, document, or window. The name of this menu matches the name of the app.
File	Contains commands to manage the files or documents the app supports. This menu may not be present if the app doesn’t handle any types of files.
Edit	Contains actions that allow you to change to content in the current document or text container, and provides commands for interacting with the Clipboard
Format	Contains actions to adjust text formatting attributes in the current document or text container. This menu may not be present if the app doesn’t support formatted text editing.
View	Contains actions that allow you to customize the appearance of the app’s window(s).
app-specific	An app may include menus for actions that are unique/specific to that app. If/when an app does this, the app-specific menus will appear between the View and Window menus.
Window	Contains actions to navigate, organize, and manage the app’s windows.
Help	Provides access to the app’s help documentation. This is always the last the menu in the menu bar.

Common Controls

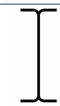
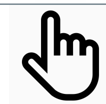
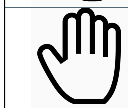
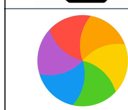
Name	Example	Description
button		Typically allows you to initiate an action; the label (i.e., text or icon inside the button) tells you what the action your device will take if/when you click it.
textbox		Allows you to enter/edit text. Some textboxes only allow you to enter a single line of text while others allow you to enter multiple lines of text. For the case of multi-line textboxes, you can usually scroll within the textbox; sometimes you can increase the size of the textbox.
checkbox		Typically allows you to enable/disable a feature or function. The box will have an “x” or a checkmark inside it if/when the checkbox is selected; the box will be empty if/when the checkbox is not selected (i.e., deslected/unselected).
radio buttons		Allows you to choose one (and only one) of a set of items. I.e., the choices presented by a group of radio buttons are mutually exclusive; when you select a radio button, any other radio button that is currently selected in the same group becomes deselected.
toggle switch		Allows you to make a choice between two mutually exclusive states—such as on/off or yes/no. Toggle switches have a similar function as checkboxes; i.e., software engineers tend to use them interchangeably when designing/building software UIs.
slider		Used to select a value from a continuous range. This control is often used to control speaker value and screen brightness. Retail sites frequently use this type of control to allow you to filter a list of products by price range.

Common Controls

Name	Example	Description
listbox	The Finder app's left-hand sidebar	Allows you to select one or more items from a list. You click on an item inside the box to select it. Some listboxes allow you to select multiple items; when they do, you press either the SHIFT or CTRL key while selecting items.
dropdown list		Similar to a listbox, this control allows you to select on or more items from a list. This control differs from a listbox in that you have to first click on the control to see the full list of options. You typically see an up arrow, a down arrow, or both up & down arrows in a dropdown list control; this arrows are indicating the control is a dropdown list. Sometimes you will need to click on the arrow(s) to see the control's list of items.
picker		Allows you to select a value from a discrete range of values. This type of control is very commonly used when selecting a date or color.
combo box		A combination of other controls—frequently list boxes, dropdown lists, and textboxes—acting in a coordinated fashion so that you have multiple ways to input/select information.
collapsible	See examples on [this web page]	Allows some content to be hidden/revealed. Software engineers use this control to reduce on screen "clutter" or conserve screen "real estate" (i.e., available sreen space).

Cursors (aka "pointers")

- A **cursor** (aka, an **arrow** or a **pointer**) is an icon used to activate or control certain elements in a user interface. The cursor indicates where your pointing device should perform its next action, such as opening an app or dragging a file to another location. The pointer follows the path of your hand as you move your pointing device.
- Moving the cursor over an object and letting it remain there for 2-3 seconds is referred to as **hovering.** Hovering over a control frequently reveals a small piece of text intended to briefly tell you what the control does.
- Further reading
    - computerhope.com: Cursor
    - geeksforgeeks.com: What is a Cursor in Computer?
    - techtarget.com: cursor
    - wikipedia.org: Cursor (user interface)

Common Cursors	
	The **pointer** cursor. This cursor is default/standard cursor. I.e., it is the cursor you will usually see. When this cursor is visible and you click or double-click with your pointing device, the object beneath the tip of the arrow is the target of the click/double-click. I.e., the object beneath the very tip of the cursor is the object you are "touching" with your pointing device.
	The **text insertion** cursor. This cursor appears when the cursor is over a control or pane in which text can be typed. When this cursor is visible, clicking your pointing device causes your device to set the text inserting point to that location in the control or pane.
	The **finger** cursor. This cursor appears when the cursor is over a link. When this cursor is visible, clicking your point device causes your device to load the page associated with the link in your browser. If you do this when using an app other than a browser, your device will load the page into your default browser. Note that you will see this cursor when you hover over some types of controls when they appear in your browser's view pane.
	The **hand** or **pan** cursor. This cursor appears when the cursor is over a pane that can be panned. This appearance of this cursor indicates the object in the pane beneath it can be moved. See the **grab** cursor below for more.
	The **grab** cursor. This cursor appears when the **hand/pan** cursor is visible and you press your pointing device's primary button. When this cursor is visible, you have "grabbed" the object on the screen beneath it; you can then move the object by moving your pointing device.
	The **busy** cursor. This cursor sometimes appears when the app you have just given a command to needs more than 2-3 seconds to complete the command. This cursor is intended to convey "please wait while we process your request."

Pointing Devices and Their Gestures

- We give our devices commands and information using keyboards, mice, trackpads, and touchscreens. In addition to the “on screen” keyboards our iPhones and iPads have, it is possible to use a wireless keyboard with iPhones and iPads as well. It is also possible to use a wireless mouse or trackpad with iPads. (I think it is also possible to use a wireless mouse/trackpad with an iPhone although the iPhone User Guide doesn’t explicitly say this is the case.)
- Mice, trackpads, and touchscreens are referred to as **pointing devices**; the physical actions we take with a mouse, trackpad, or touchscreen are called **gestures**. All pointing devices support a relatively small set of basic/core gestures; some pointing devices support additional gestures beyond this basic/core set. Regardless, mastery of the basic/core set of gestures is key to effectively using our digital devices.
- This document describes these basic/core gestures, how to perform them, and how they correspond to each other across pointing devices. Note that a gesture does often depends on the context in which you perform the gesture. We will see examples of these context dependencies as we explore our devices’ **user interfaces** (**UIs**) more broadly.
- This document also introduces **keyboard shortcuts**. Keyboard shortcuts are a way to give your device commands (i.e., actions you want it to perform). Using a keyboard shortcut is usually an *alternative* to using a menu item, icon, or button on your screen although in a few cases (e.g., taking a screenshot on a laptop), keyboard shortcuts are the *only* way to tell your laptop to perform an action.

Mouse Basics

Computer mice have at least three buttons. Many (most?) mice have two buttons and a scroll wheel. The scroll wheel acts as the third button; i.e., you can press the scroll wheel to use it as a button.) Familiarity with the two standard buttons all mice have is key to using a mouse.

The image to the right is of a simple/basic mouse. This mouse has an "ambidextrous design." I.e., it can be used with either the right or the left hand. Some mice are designed to only be used with one hand (usually the right since most people are right-handed.)

When you see"**click** on" in a user/help guide or hear someone say "click on" something, it means use the primary mouse button. When you see/hear "right-click on" or "alt-click on", it means use the secondary mouse button.

To perform a **click** with a mouse, you press and immediately release the button *without* moving the mouse. Performing a **double-click** is simply a matter of **clicking** twice. **Double-clicking** is always done with the primary button—never with the secondary button.

When you see "**press**" in a user/help guide or hear someone say "press on", it means to depress the primary button and keep it pressed *while* you move the mouse.

You can customize some mouse behaviors via some system settings on macOS: > System Settings > Mouse.

Buttons

Every mouse's buttons are found at the front/top of the mouse. The button nearest your body is the **primary button;** the button furthest from your body is the **secondary button.** If you use your mouse with your right hand, the primary button is on the left-hand side of the mouse and the secondary button is on the right-hand side. Alternatively, if you use your mouse with your left hand, the **primary button** is on the right-hand side of the mouse, and the **secondary button** is on the left-hand side

In user guides and help documentation, the primary button is often referred to as the **left-click button** while the secondary button is often referred to as the **right-click button.** If you use a mouse with your left hand, keep in mind that the left-click button is the button on the right-hand side of your mouse and the right-click button is on the left-hand side.

Scroll wheel

Most mice sold today also have a scroll wheel between the two basic buttons; you can both roll the scroll (like a dial) and click the scroll wheel (like a button). When using the scroll wheel like a button, it is referred to the **middle** button.

The few mice sold today that do not have a scroll wheel typically have just a button in this spot.

Comparison of Key Pointing Device Gestures

Mouse	Trackpad	Touchscreen	Description
click	tap	tap	Arguably the most common gesture you use. This gesture typically causes the object beneath the cursor to (1) act (i.e., perform the action associated with the object beneath the cursor), (2) highlight/select the object beneath the cursor, or (3) set the insertion point when performed over a textbox or document pane.
double-click	double-tap	*not supported*	This gesture is most commonly used to 1. select an object (where the object is text or an image in a document or on a web page) 2. open an object (normally a file or app)
alt-click *aka right-click*	two-finger tap	press	The meaning/purpose of this gesture varies more between a touchscreen and a mouse/trackpad. Performing this gesture on your laptop—with either a mouse or a trackpad—causes a context menu to appear. On a touchscreen, this gesture is more “context sensitive”; sometimes it causes a context menu appear; other times it sets the insertion point or selects the object you press on.
press	press	press	The meaning purpose of this gesture is somewhat “context sensitive”. On a laptop, this gesture is used to “pick up” an object when performing a drag-and-drop operation, but it does other things in other contexts. On an iOS device, this gesture tends to cause a context menu to appear.
drag	drag	drag	On a laptop, this gesture is performed after pressing on an object by • moving the mouse after pressing on an object • sliding your finger across the trackpad • sliding your finger across the touchscreen

Note: Apple trackpads are relatively highly configurable via system settings (> Settings > Trackpad).

For example,

- Trackpad settings can be adjusted so that the “tap” gesture is either literally done by tapping a finger on the touchpad or placing a finger on the trackpad and pressing until you feel a “click”.
- Instead of using a “two-finger tap” to perform an “alt-click”, you can change your trackpad’s settings such that you click the bottom right corner of the bottom left corner of your trackpad instead.

Pointing Device Help Guide References

- Mac User Guide
  - View and customize mouse or trackpad gestures on Mac
  - Use Multi-Touch gestures on Mac
  - Right-click on Mac
- MacBook Air Getting Started Guide: MacBook Air trackpad
- MacBook Pro Getting Started Guide: MacBook Pro trackpad
- iPhone User Guide
  - Learn basic gestures to interact with iPhone
  - Learn gestures for iPhone models with Face ID
- iPad User Guide
  - Learn basic gestures to interact with iPad
  - Learn advanced gestures to interact with iPad
  - Move and copy items with drag and drop on iPad
- See the Accessories section of the iPad User Guide for more information on using an external keyboard, mouse, and/or trackpad with an iPad
- See the Accessories section of the iPhone User Guide for more information on using an external keyboard with an iPhone

Keyboard Shortcuts

A **keyboard shortcut** is a *sequence* of one or more keys on the keyboard that can be used to perform an app's menu action or an operating system command. Keyboard shortcuts are also called **hot keys**, **shortcut keys**, or sometimes simply a **shortcut.**

When a shortcut involves more than one key, you press and hold each of the keys in the order listed. With the exception of the last key in the sequence; you just have to tap the last key in the sequence. Once you have tapped that last key, the associated action will execute/run. Also, once you have tapped the last key in the sequence, you can then release all the keys you have pressed.

In help guide documentation, keyboard shortcuts are written as the sequence of keys you use separated by the "+" (plus) sign. For example:

- CMD+N *or* ⌘+N *means "press the command key and then tap the N key"*
- SHIFT+CMD+Z *or* SHIFT+⌘+Z *means "press the shift key, then press the command key, then tap the Z key"*

If/when a menu action has a shortcut, you will usually see the shortcut's key sequence to the right of the menu action in the menu where that action appears.

The following keys are often used in keyboard shortcuts:

Name	Symbol
control	CRTL *or* CTL *or* ^ (caret)
command	CMD *or* ⌘
alt *or* option	ALT *or* OPT *or* ⌥
shift	SHIFT
spacebar	Space
function	fn
escape	esc

The table below lists the most "popular" actions/commands that have shortcuts. Note that the first four actions/commands are available in any/all apps that allow you to create/save/open files. The other actions/commands tend to be available at all times—i.e., when using any app.

Action	Shortcut
New (file/document)	CMD+N
Open (file/document)	CMD+O
Save (file/document)	CMD+S
Close (window)	CMD+W
Quit (app)	CMD+Q
Copy	CMD+C
Cut	CMD+X
Paste	CMD+Paste
Undo	CMD+Z
Redo	SHIFT+CMD+Z
(Open the) Task Switcher	CMD+Tab
(Open) Spotlight Search	CMD+Space